

witt-et-al-2023-variability-of-dot-spread-is-overestimated

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Paper Synthesis

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Abstract

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Previous research has demonstrated that individuals exhibit a tendency to overestimate the variability of both low-level features (e.g., color, orientation) and mid-level features (e.g., size) when items are presented dynamically in a sequential order, a finding we will refer to as the variability overestimation effect. Because previous research on this bias used sequential displays, an open question is whether the effect is due to a memory-related bias or a vision-related bias. To assess whether the bias would also be apparent with static, simultaneous displays, and to examine whether the bias generalizes to spatial properties, we tested participants' perception of the variability of a cluster of dots. Results showed a consistent overestimation bias: Participants judged the dots as being more spread than they actually were. The variability overestimation effect was observed when there were 10 or 20 dots but not when there were 50 dots. Taken together, the results of the current study contribute to the ensemble perception literature by providing evidence that simultaneously presented stimuli are also susceptible to the variability overestimation effect. The use of static displays further demonstrates that this bias is present in both dynamic and static contexts, suggesting an inherent bias existent in the human visual system. A potential theoretical account--boundary effect--is discussed as a potential underlying mechanism. Moreover, the present study has implications for common visual tasks carried out in real-world scenarios, such as a radiologist

making judgments about distribution of calcification in breast cancer diagnoses.

Keywords Ensemble perception · Visual biases · Perceived spread · Variability

A critical function of human perception is the ability to accurately process and perceive variability in our environment. People are continually confronted by variability of all forms, as objects can differ along several dimensions including color, size, weight, speed, and orientation. As a consequence, humans have developed the ability to rapidly extract summary statistical information from groups of items, a process known as ensemble perception (for review, see Whitney & Yamanashi Leib, 2018). For example, people can see the mean orientation, color, and size of objects, and even the mean emotional expression of a group of faces. People can also detect other summary statistics such as the variability of a set of items (Dakin & Watt, 1997; Haberman et al., 2015; Morgan et al., 2008; Norman et al., 2015; Solomon et al., 2011).

And yet, despite the incredible capacity for ensemble perception, biases also exist. For example, perceivers underestimate variability when stimuli are presented in an ordered fashion (e.g., smallest to largest) relative to when the order is randomized (Lathrop, 1967). Perceivers also underestimate the variability when line-drawings of cylinders are colored to various heights (Kareev et al., 2002). Increased variability also leads to a bias to overestimate the mean size of circles and mean length of lines (Semizer & Boduroglu, 2021). Moreover, Kanaya et al. (2018) have reported amplification effects wherein viewers' perception of the mean size and temporal frequency of array are biased by the most salient items in the display, and that this effect increases as the number of items in the salient group increase.

Most relevant to the current work, a recent bias to overestimate variability has been observed in a series of experiments in which items were presented dynamically (Witt, 2019). In one experiment, participants viewed a set of nine lines, each presented at a different orientation and in a rapid sequence. Participants estimated the variability of the set of lines by performing a bisection task for which they selected one of several response options with some options being low in variability and other options being high in variability. Bisection tasks have been used to measure time perception in pigeons, mice, humans, and rats (Penney et al., 2008; Raslear, 1985) and speed perception in humans (Witt & Sugovic, 2010).

These tasks require observers to form an implicit midpoint, and the measurement of this midpoint is compared to the objective midpoint to identify the direction and magnitude of any bias. When judging the variability of line orientation, participants overestimated the variability of the lines by approximately 50% (Witt, 2019). We refer to this bias to overestimate the variability of the ensemble as the variability overestimation effect.

The variability overestimation effect is not limited to line orientation. Follow-up studies have documented a similar bias to overestimate variability of circle size and color value (Warden et al., 2022; Witt et al., 2019). In one set of studies, participants saw a set of circles that varied in

size and estimated variability by performing the bisection task for which they judged whether the variability of the circle sizes was "less" or "more" variable. In another set of studies, participants saw circles that varied in color value (such as shades of green) and judged whether the variability of the circles' color values were "less" or "more" variable. Participants overestimated the variability for both features. This reveals the variability overestimation effect generalizes to multiple features including low-level features of orientation and color value and the mid-level feature of size.

To date, however, the research on the variability overestimation effect has exclusively used dynamic, sequentially presented stimuli. Each item in a set was presented sequentially with only one object in the set visible at a time and each item presented in the same spatial location. This raises the issue of whether the variability overestimation effect is specific to sequential displays or whether the bias would also be present when all items in the set are presented simultaneously. This issue is relevant for mechanistic and for applied reasons. If the variability overestimation effect is observed only with sequential displays, this could be indicative that the effect is due to a memory-related bias. In contrast, if the variability overestimation effect is also observed with simultaneous displays, this could be indicative of a general bias inherent in the visual system that could impact many if not all forms of ensemble perception.

With simultaneous displays, it is also possible to determine whether the effect is observed when variability is defined in terms of the spatial properties between items. In the same way that we can rapidly extract information about the mean size, color, or orientation of items, there is considerable evidence that ensemble information can be rapidly extracted from spatial arrays that are not necessarily defined by featural differences. For example, using the centroid paradigm (Drew et al., 2010; Sun et al., 2016, 2018), it has been shown that individuals are able to accurately determine the centroid (e.g., center of mass) of a briefly presented array of dots. Accurate centroid perception has been reported with a variety of different display types and tasks (e.g., Friedenbergs & Liby, 2002; McGowan et al., 1998).

Biases have also been reported, however, which influence the accuracy of spatial perception in these tasks. As one example, dots clustered more closely together exerted less weight on estimated mean position (i.e., centroid) compared with more isolated dots (Rashid & Chubb, 2021). As another example of bias, observers were not as accurate in determining the centroid of an array of target stimuli when presented simultaneously with distracting stimuli that were to-be-ignored (Drew et al., 2010) and were even worse when the target stimuli are comprised of two or more levels of a single feature dimension (Lu et al., 2019). Adaptation effects also lead to biases: adaptation to dense visual texture affects accurate perception of both the numerosity and spatial distribution of stimuli comprising a texture (Durgin, 1995).

In the current experiments, we tested whether the variability overestimation effect is also present when viewing a cluster of dots. These studies test the generalizability of the variability

overestimation effect by testing its presence with simultaneous displays and with an ensemble that varies along the spatial property of location or position, rather than object properties such as orientation, color value, and size. Testing the generalizability of the bias helps determine the boundary conditions of the effect, which has implications for the underlying mechanisms such as whether there is likely to be a memory-related component.

From a practical standpoint, the variability overestimation effect has real-world implications. There are many instances for which it is necessary to judge the variability of an ensemble presented simultaneously. Within the realm of data visualization, the variability of the data can be an important part of the message being communicated, so biases in perception of the variability could impact data interpretation. Szafir and colleagues (Albers et al., 2014; Szafir et al., 2016) have shown that individuals can make judgments about the aggregate properties of data presented to them in single images and that their ability to extract certain types of information is a function of the visualization design: different display types lead to differential efficiency in ensemble processing.

A real-world example with life-or-death consequences is mammography. To diagnose breast cancer, radiologists must perform the visual task of identifying cancer in a scan. One of these visual tasks is to classify the distribution of calcifications in the breast into one of five categories (see Fig. 1a). As this is primarily a visual task, biases within the visual system could impact diagnostic accuracy. In the present study, we examined whether the variability overestimation effect is also present in a task in which all relevant stimuli are presented simultaneously. To give the task real-world relevance, the displays were designed to mimic one aspect of the task of a radiologist: Participants viewed an ensemble of dots that were either more clustered or more dispersed and judged the spread of the dots (see Fig. 1b).

Fig. 1 a Examples of the five categories of calcification distributions for diagnosing breast cancer (image courtesy of Dr. Elizabeth Edney; Witt et al., 2022), and (b) examples of our experimental stimuli. In (b), the first image shows the lowest level of spread with increasing levels for each subsequent image, and the last image shows the highest level of spread

If the variability overestimation effect (such as with line orientation) is also found in the perceived variability of simultaneously presented dots, this could have implications for accurately diagnosing breast cancer. A bias to overestimate variability in calcification spread could lead to distributions that should be classified as "clustered" as being "diffuse." Given that diffuse distributions portend a benign diagnosis, a visual bias to overestimate variability could lead to a higher rate of false negative diagnoses (i.e., "misses"). These misdiagnoses would lead to delays and missed opportunities for easier, more effective early treatments. To the extent that any pathological condition depends on a visual judgment of a distribution, the bias to

overestimate variability in the distribution could have adverse consequences.

Introduction (Paper-Level)

Not available for this paper.

General Discussion

Not available for this paper.

Critique

Not available for this paper.